

SUDATTA RAY

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ACADEMIC APPOINTMENTS

Assistant Professor	National University of Singapore (NUS) Department of Geography	August 2022 -
Assistant Professor (Joint Appointment)	Yale-NUS College Department of Environmental Studies	January 2022 -

EDUCATION

PhD	Stanford University Emmett Interdisciplinary Program in Environment and Resources (E-IPER) Dissertation: Impact of Rural Electrification on Food Security of Smallholder Farmers in India	2021
MSPPM	Carnegie Mellon University H. J. Heinz III College, School of Public Policy and Management Master of Science in Public Policy and Management – Data Analytics Track	2014
MS	University of Delhi St. Stephen's College Master of Science in Chemistry	2009
BS	University of Delhi Sri Venkateswara College Bachelor of Science in Chemistry	2007

PUBLICATIONS

PUBLICATIONS

Ray, S., Pullabhotla, H.K. 2023. "[The changing impact of rural electrification on Indian agriculture.](#)" *Nat Commun* **14**, 6780 (2023).

Ray, S., Benson, S. 2024. "[Energy models in service of aquifer specific groundwater irrigation expansion in India](#)" *Environ. Res. Lett.*, **19** 104043.

WORKS IN PROGRESS

Ray, S. & Gupta I. "Cascading effects of electrification on water and sanitation among marginalized communities."

Ray, S., Falcon W. & Naylor, R. "Rural Electrification in Agricultural Intensification Across India."

Ray, S. "Split or aligned incentives? The case of plastic recycling vs incineration in Southeast Asia."

Ray, S. "Measuring uncertainty in global Greenhouse Gas reporting for biomass energy."

Ray, S. "More nutrients per drop of groundwater: drivers of vegetable cultivation and consumption in India."

POLICY REPORTS

Jain, A., **Ray, S.**, Ganesan, K., Aklin, M., Cheng, C.& Urpelainen J. (2015). "[Access to Clean Cooking Energy and Electricity, Survey of States.](#)" *Columbia University and Council on Energy, Environment and Water Report.*

Ray, S., Chaturvedi, V., Ganesan, K. & Ghosh, A. (2015). "[India's Intended Nationally Determined Contributions: Renewable Energy and Pathway to Paris.](#)" *Council on Energy, Environment and Water Policy Brief.*

Ghosh, A. & **Ray, S.** (2015). "[Fixing Climate Governance through Effective Technology Partnerships.](#)" *Fixing Climate Governance Series, Paper No. 3.* Centre for International Governance Innovation.

Ghosh, A., Vijayakumar, A. & **Ray, S.** (2015). "[Climate Technology Partnerships: Form, Function and Impact.](#)" *Fixing Climate Governance Series, Paper No. 2.* Centre for International Governance Innovation.

Ray S. (2015). "[Electricity Trading in its Infancy in South Asia.](#)" *India Climate Dialogue Web Article.*

Ganesan, K., Jain, A., **Ray, S.**, Sharma, M. & Ghosh, A. (2014). "[Agenda for a Reformed Power Sector in India: Risk, Resource, Relay and Restructuring.](#)" *Council on Energy, Environment and Water Policy Brief.*

[Black Carbon Research Initiative](#) – *National Carbonaceous Aerosols Programme (NCAP)*, Ministry of Environment & Forests, March 2011 – Editorial Team.

[Climate Change & India](#) - *A 4x4 Assessment, A Sectoral and Regional Analysis for 2030s*, Ministry of Environment & Forests, November 2010 – Editorial Team.

[India: Greenhouse Gas Emissions 2007](#), Ministry of Environment & Forests, May 2010 – Editorial Team.

PRESENTATIONS & CONFERENCE ACTIVITY

CONFERENCE ACTIVITY

Ray, S. (2024). "Cascading effects of spatial segregation and energy poverty on groundwater access." *35th International Geographical Congress, 2024.* Presentation, (Dublin, Ireland).

Ray, S. (2024). "Cascading effects of spatial segregation and energy poverty on groundwater access." *Royal Geographical Society with the Institute of British Geographers (RGS-IBG), Annual International Conference, 2024.* Presentation, (London, UK).

Ray, S. (2024). "Balancing groundwater access and sustainability through energy pricing." *European Geosciences Union, General Assembly 2024.* Presentation, (Vienna, Austria).

Ray, S. (2023). "Balancing groundwater access and sustainability through energy pricing." *American Geophysical Union, Fall Meeting 2023.* Poster, (San Francisco, USA).

Ray, S. (2020). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Association for Public Policy Analysis & Management (APPAM).* Presentation, (Online due to COVID).

Ray, S. (2020). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Sustainable Energy Transition Initiative (SETI) Annual Conference, Duke University.* Presentation, (Online due to COVID).

Ray, S. (2020). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Agricultural & Applied Economics Association, Annual Meeting.* Presentation, (Kansas City, USA-Online due to COVID).

Ray, S. (2020). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Online Summer Workshop in Environment, Energy, and Transportation (Economics).* Presentation, (Online).

Ray, S. (2020). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Association of Environmental and Resource Economists, Annual Summer Conference.* Presentation, (Miami, USA-Online due to COVID).

Ray, S. (2019). "Rural Electrification in Agricultural Intensification Across India." *American Geophysical Union, Fall Meeting 2019.* Poster, (San Francisco, USA).

Ray, S. (2019). "The Evolving Role of Rural Electrification in Indian Agriculture." *The University of Michigan Ann Arbor, Sustainability & Development Conference 2019.* Presentation, (Michigan, USA).

- Ray, S. (2018). "Beyond Tube-wells: Bringing Irrigation to Oriya Smallholder Farmers." *IWMI- Tata's Partners' Meet 108, Building Climate Resilience for Doubling Farmers' Income – Innovations in Water, Land, Energy and Ecosystems*. Presentation, (Gujarat, India).
- Ray, S. (2015). "Analysing Climate Actions and their Global Consequences." *World Resources Institute (WRI) Technical Practitioner's Workshop for Analyzing Mitigation INDCs*. Presentation, (Bonn, Germany).
- Ray, S. (2015). "India's INDC: Renewable Energy and the Pathway to Paris." *Council of Energy, Environment and Water (CEEW) Climate Day, Negotiating the Climate Cliff*. Presentation, (New Delhi, India).

INVITED TALKS

- Ray S. (2024). "Re-imagining Rural Electrification Beyond Lights." *Asian School of Environment & Earth Observatory of Singapore Seminar Series (AES-EOS)*, Nanyang Technological University, Singapore.
- Ray S. (2023). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *Asia Research Institute*, National University of Singapore, Singapore.
- Ray S. (2023). "Beyond Lights: The Changing Impact of Rural Electrification in Indian Agriculture." *FASS Brown Bag Seminar Series*, National University of Singapore, Singapore.
- Ray, S. (2020). "Seeing the Light: Harnessing Rural Electrification to Boost Development" *Hard Earth Lecture Series*. Stanford University, California.
- Ray, S. (2019). "Energy Access for Development: Scope and Need for Productive Uses of Electricity." *Global Climate and Energy Project, Student Lecture Series*. Stanford University, California.
- Ray, S. (2019). "State of Energy Access in India." *Energy Access Panel, Stanford Energy Club*. Stanford University, California.
- Ray, S. (2016). "Impact of Energy Policies on Resources Use and Food Security in India." *Enhancing Diversity in Graduate Education, Cross Talks*. Stanford University, California.

TEACHING

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|---|--------------|
| Environmental Studies Division, Yale-NUS College | 2022-Present |
| <ul style="list-style-type: none"> - Powering the Planet (2022,2023). - Environmental Policy Analysis (2024). | |
| Geography Department, National University of Singapore | 2022-Present |
| <ul style="list-style-type: none"> - Applied Geographical Data Science (2023, 2024). - Just and Clean Energy Transition (Winter, 2025). | |

ADVISING

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| Environmental Studies Division, Yale-NUS College | 2022-Present |
| <ul style="list-style-type: none"> - Nageen Ayesha Rameez (2023-24) – Honors Thesis, Outstanding Capstone Award - Rinko Kawamoto (2023-24) – Honors Thesis - Aye Chan Phyu Sin (2023-24) – Independent Research Module - Joseph Goh (2023 Summer) – Summer Research Programme - Amanda Chen (2022-23) – Honors Thesis | |
| Geography Department, National University of Singapore | 2022-Present |
| <ul style="list-style-type: none"> - Baoyan Han (2024 – Present) – Secondary Adviser, Doctoral Committee - Siyuan Kang (2024-25) – PhD Independent Study Module (ISM) | |

- Fadhlina Suhaimi (2022-23) – Masters Independent Study Module (ISM)

RESEARCH EXPERIENCE

- Graduate Research Student and Instructor, **Stanford University** 2015-2021
- Estimated the impact of rural electrification policy change on the growth in groundwater irrigation between 1986 and 2013 for 323 districts in India.
 - Modeled the impacts of different energy provision schemes on smallholder farmer income, groundwater sustainability and carbon emissions for 30 districts in Orissa, India.
 - Surveyed and analysed electricity use for 500 households across 5 urban informal settlements in Harare, Zimbabwe.
 - Taught "Capstone Project Seminar in Environment and Resources" (28 students across 2 years).
 - Developed course curriculum and taught "Achieving Universal Energy Access by 2030: Can it be Done?" (15 students).
- Junior Research Associate, **Council on Energy, Environment and Water (CEEW)** 2014-2015
- Surveyed and analysed the state of energy access in 8,566 rural households across 6 states in India.
 - Developed integrated plans in agriculture, energy and water for tribal districts in Madhya Pradesh, India.
 - Analysed and recommended steps for reforming the Indian power sector to the Office of the Prime Minister of India.
 - Estimated the financial implications of India's Intended Nationally Determined Contribution to the UNFCCC.
- Climate Corp Fellow, **Environmental Defense Fund** 2013 Summer
- Developed energy efficiency projects for Maricopa County Community Colleges and facilitated annual cost savings of \$220,000 and GHG emissions savings of 1,520 CO₂ eq.

POLICY EXPERIENCE

- Senior Programme Officer, **World Wide Fund for Nature – India** 2011-2012
- Led analyses of international climate change negotiations as part of the Global Climate Energy Initiative.
 - Analysed the State Action Plans on Climate Change for five Indian Himalayan states.
 - Developed a framework to track the efficacy of low carbon policies of ten states across India.
- Technical Officer, **Government of India, Ministry of Environment, Forests and Climate Change** 2009-2011
- Negotiated India's stand at 2 Conference of Parties and 2 Intersessional Meetings of United Nations Framework for Climate Change Convention (UNFCCC) Climate Change talks on issues of HFC mitigation and market mechanism.
 - Wrote briefs and responses to climate change related questions raised in the parliament.
 - Led a nation-wide initiative for the development of State-level Action Plans on Climate Change.

GRANTS

Yale-NUS Seed Grant (SGD 68,500)	2024
Yale-NUS Start-up Grant (SGD 75,000)	2022
EIPER Collaboration Research Grant (\$10,000)	2019
McGee Grant (\$3,000)	2018
EIPER Summer Graduate Research Grant (2,110)	2018
EIPER Summer Graduate Research Grant (\$2,500)	2017
McGee Grant (\$3,790)	2016
Enhancing Diversity in Graduate Education (EDGE) Travel Grant (\$863)	2015

FELLOWSHIPS & AWARDS

School of Earth, Energy and Environmental Sciences Certificate for Outstanding Mentoring	2020
Kimmelman Family Fellowship	2015- 2019
Enhancing Diversity in Graduate Education Fellowship (EDGE-STEM) Fellow	2015-2017
Fulbright- Nehru Master's Fellowship in Leadership and Development	2012-2014
HEINZ MSPPM Merit Based Scholarship	2012-2014
Second Prize, Annual Social Innovations Challenge, Carnegie Mellon University	2013

DEPARTMENT & UNIVERSITY SERVICE

Graduate Student Committee	2023- Present
Reviewer for <i>Environmental Research Letters</i> , <i>Earth's Future</i> , <i>Journal of Environmental Studies and Sciences</i> , <i>Singapore Journal of Tropical Geography</i>	
Enhancing Diversity in Graduate Education (EDGE) Mentor	2017-2018
E-IPER Admissions Committee	2017